

## rStrcmp

The **recursive** C function that compares the string pointed to by *s1* to the string pointed to by *s2*. If the string pointed to by *s1* is greater than, equal to, or less than the string pointed to by *s2*, then it returns 1, 0 or -1 respectively. Write the code for the function without using any of the standard string library functions. The function prototype is given as follows:

```
int rStrcmp(char *s1, char *s2);
```

A sample program template is given below to test the function:

```
#include <stdio.h>
#include <string.h>
#define INIT_VALUE 100
int rStrcmp(char *s1, char *s2);
int main()
{
    char source[40], target[40], *p;
    int result = INIT_VALUE;

    printf("Enter a source string: \n");
    fgets(source, 40, stdin);
    if (p=strchr(source, '\n')) *p = '\0';
    printf("Enter a target string: \n");
    fgets(target, 40, stdin);
    if (p=strchr(target, '\n')) *p = '\0';
    result = rStrcmp(source, target);
    printf("rStrcmp(): %d", result);
    return 0;
}
int rStrcmp(char *s1, char *s2)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

- (1) Test Case 1:  
Enter a source string:  
abc  
Enter a target string:  
abc  
rStrcmp(): 0
- (2) Test Case 2:  
Enter a source string:  
abcdef  
Enter a target string:  
abc123  
rStrcmp(): 1
- (3) Test Case 3:  
Enter a source string:  
abc1234  
Enter a target string:

```
abc123  
rStrcmp(): 1
```

(4) Test Case 4:  
Enter a source string:  
abc123  
Enter a target string:  
abc123f  
rStrcmp(): -1